Agentic delta proofs

Landing many agents' work on one trunk without a queue, and what lung has of it

Version 1. Design note with the working rules (appendix A), the lander sketch with review notes (appendix B), and where lung stands.

CONTENTS

1 The problem 2
2 Two ways to land 2
3 The delta proof 3
- 3.1 Repeatable results 4
- 3.2 When the check fails 4
- 3.3 Changes that do not collide 4

4 Working rules for agents 5
5 Keeping main healthy 5
6 Rollout in four phases 6
7 Going further: machine-checked proofs 6
8 Applied to lung 7
9 Appendix A: the AGENTS.md text 10
10 Appendix B: the lander sketch, with review notes 11

THE SHORT VERSION

- **The trouble.** With dozens of agents changing one codebase at once, the old routine (a branch, a full test run, a wait in line) falls behind. Branches go stale, pushes bounce, and everyone rebases again and again.
- **The idea.** Let each worker run the full tests on its own machine and attach a signed-off note, a *certificate*, saying which version of main it tested against. The lander then does only a small check. If main has not moved, it lands the change at once. If main has moved, it runs only the tests that touch what moved.
- **The safety net.** One writer updates main, one change at a time. If a landed change breaks the later full checks, main freezes and an automatic revert puts it right.
- **Lung, as measured.** Lung already has the plumbing (a worktree per task, one lander at a time, two tiers of checks, a revert watchdog). It has no certificate, no overlap check, and no working freeze. Main is red on most recent pushes for a reason that is not the code, and that comes first (page: Applied to lung).

1 The problem

When agents work without a person reviewing each change, main is the shared memory of the whole swarm. Every agent builds its picture of the code from a snapshot of main. The longer a branch lives, the further that snapshot is from the real thing, and three things go wrong.

1. **Drift.** The amount of change a branch has to catch up on is the total of everything landed on main since it began. The more commits land per hour, the faster a branch goes stale, and the more an agent has to read just to see what changed.
2. **Stale pictures.** An agent keeps writing against functions and types that another agent has already changed or deleted.
3. **Quiet conflicts.** Two changes can merge without a text conflict and still break each other, for instance by changing what a function promises or the order of steps. Nothing flags it until something fails.

The usual pipeline makes this worse. It runs the whole test suite, in turn, for every change. With many agents landing at once the line grows, branches fall behind while they wait, pushes are refused because main moved, and agents rebase again. Work churns without finishing.

2 Two ways to land

There are two designs that stop the line from collapsing. The choice depends on how many changes land per day and how long the checks take.

QUESTION	A: MERGE QUEUE	B: ONE WRITER, STRAIGHT TO MAIN
How much work	Up to about 60 commits a day	More than 60 a day
How fast the checks must be	Whole check in 5 minutes or less (2.5 under load)	A pre-check of 60 seconds or less, then full checks after landing
How a change lands	A short-lived branch goes into an automatic queue that tries it against current main and lands it	The worker rebases in its own copy, a single lander takes the lock and pushes to main
Record of what happened	The pull request history	Signed commits and an append-only log
How it fails	Line builds up, rebases repeat	A bad change reaches main briefly and must be undone fast

Table 1. The two landing designs. Source: the design note, section 2.

Design A fits when the load is modest, checks are quick, and changes often break each other. Design B fits when volume is high and checks run longer than 2.5 minutes, because a one-at-a-time queue then cannot keep up. B needs the after-landing watchdog described below.

3 The delta proof

The aim is to keep the slow work (the full tests) away from the one step that must happen in single file (the write to main). Workers do the slow work, in parallel, in their own copies. The lander only holds the lock for a quick update.

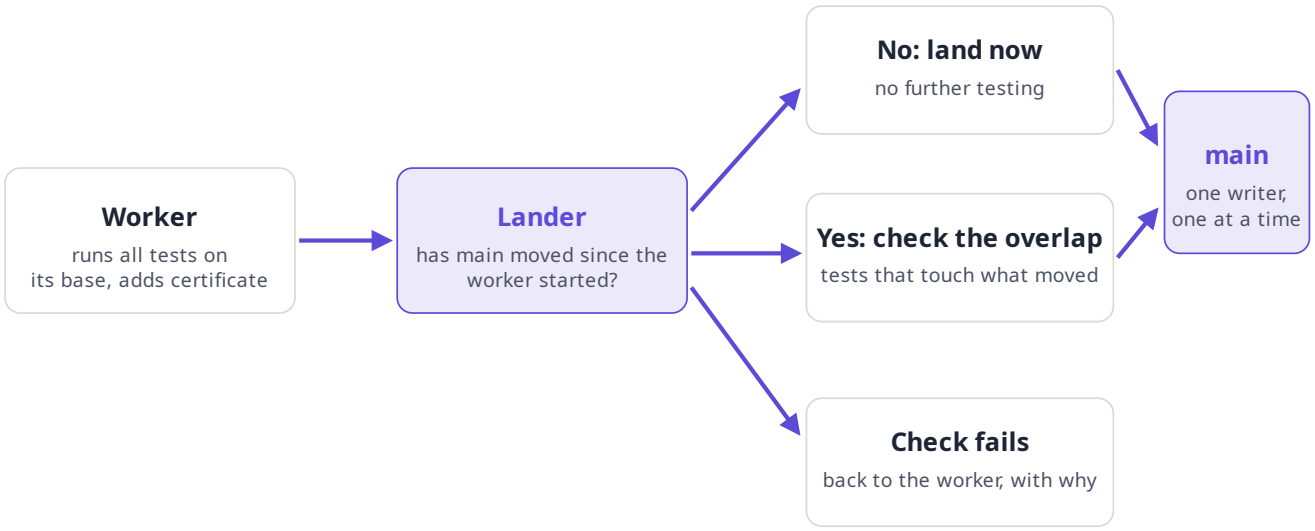


Figure 1. The lander flow. The worker brings a certificate; the lander compares it with where main is now and takes the short road when it can.

A worker runs the full tests on the version of main it started from, call it C, and records the result in a certificate. The lander rejects any submission without a valid certificate for the version it claims.

FIELD	WHAT IT HOLDS
Submission id	A unique number for this attempt
Base	The exact version of main the worker started from (40 characters)
Diff hash	A fingerprint of the change, so the certificate cannot be moved to a different change
Suite status	Pass or fail
Affected contracts	The parts of the code whose promises this change touches

Table 2. The certificate. It travels as lines in the commit message or in a side record in the repository.

The lander then takes one of two roads.

1. **Nothing has landed since C.** Main is still exactly where the worker tested, so the change is landed at once with no more testing.
2. **Something has landed since C.** The lander tries the two together in a scratch copy and runs only the tests that touch both the submitted change and what moved, plus the checks for repeatable results. If no tests touch both, this takes seconds.

The work of the lander is therefore bounded by how far main has moved, not by the size of the whole codebase.

3.1 Repeatable results

A certificate is only worth something if the tests give the same answer every time. Before certifying, the worker runs checks that catch the usual causes of different answers: listing files or items in whatever order the system gives them (always sort), unordered key lists, and searches whose results arrive in unpredictable order. If a result ever differs between runs, the certificate is void and the landing is refused.

3.2 When the check fails

The lander stops, and sends the exact failure back to the worker that submitted. The worker rebases in its own copy, fixes the collision, and gets a fresh certificate. The effort returns to the many workers instead of piling up at the centre.

3.3 Changes that do not collide

If two changes touch different functions or different modules, applying them in either order gives the same result. So they can be handled together and landed in any order without voiding each other’s certificates. Where they would touch the same thing, a reservation made before the work starts (a lock on that file or function) keeps the second worker waiting instead of colliding.

4 Working rules for agents

The design relies on small, safe changes. These are the habits it asks of every worker. The full text is appendix A.

RULE	DETAIL
Own copy of the code	Each task works in its own worktree, cut from the current main
Time limit	A worktree with no passing small change after 30 to 45 minutes is thrown away and the task is split
Small changes	At most 150 lines of real code per commit; larger work is split in steps
Switches default off	New behaviour is behind a switch that starts off, so unfinished work can land safely
Replace in steps	Add the new next to the old, move callers over a few at a time, then delete the old
Rename in steps	Never drop or rename a database field in one go: add, then move readers and writers, then remove
Two tiers of checks	Tier 1 before landing, under 60 seconds. Tier 2 after landing, under 5 minutes
Standard commit notes	Conventional commit titles with fixed lines (Task-ID, Agent-Run, Feature-Flag, Certificate-Base) a script can read

Table 3. Working rules. Source: the design note, section 4.

Tier 1 holds the type check, a warnings-as-errors build, the tests affected by the change, and the repeatability checks. Tier 2 holds the full integration tests, fuzzing, speed checks, and a watchdog that keeps testing the head of main.

5 Keeping main healthy

If Tier 2 fails on main, main is red. While it is red:

- The lander refuses all ordinary submissions, so nobody builds on a bad base.
- It accepts only a revert or a designated hotfix.
- A watchdog reverts the failing commit straight away, then files a diagnostic task for a repair agent to work on in a clean copy.

Agents may not try to fix forward on a broken main. They pull the revert, reproduce the problem in a fresh worktree, and submit again with a new certificate.

Under design B, workers may not push to main directly. They send the commit and certificate to a single coordinator, which rebases, checks, and pushes inside a short lock, normally tens of milliseconds.

6 Rollout in four phases

PHASE	WEEKS	WHAT LANDS
1. Agree the protocol	1 to 2	The root AGENTS.md, worktree per task, time limit, small commits, repeatability checks, local Tier 1
2. Certificates and the lander	3 to 4	Certificates from the worker, the lander with its empty-change shortcut, failures sent back to workers
3. Guardrails	5 to 6	Freeze on red through branch protection, the Tier 2 watchdog, standard commit lines
4. Many writers	7 and on	Reservations on functions or files, parallel checks for unrelated changes, then the single coordinator

Table 4. The rollout. Source: the design note, section 7.

7 Going further: machine-checked proofs

For safety-critical code, the certificate can be a mathematical proof checked by the Lean 4 proof assistant in place of a test log. The worker proves that its change keeps the promises of each interface; the lander checks the proof without re-reading the code. When the changes that landed in between leave those promises alone, the lander accepts the combination from the proofs alone. The lander’s own rules can also be written in Lean, so its decisions are checked by a kernel and not left to shell scripts, and unordered iteration can be made impossible to write at all.

8 Applied to lung

Sources: a read-only study of lung at main version 97f75f0, which changed no repository.

In one line. Lung has the plumbing of this design but not its proof objects. It has a worktree per task, one lander at a time, two tiers, ratchets with self-tests and a revert watchdog. It has no certificate, so the lander runs the whole of Tier 1 for every change. It has no overlap check, no way to make edits commute (44% of changes that are open at once touch the same file), a freeze that has not landed, and landing decisions made in shell, not in the spec.

Do first. Main is red on 24 of the last 40 pushes, because of the way Tier 2 checks out the code and not because of any commit. Until that is fixed, a working freeze would freeze lung for good.

MEASURE	VALUE
Commits on main per day (Nov 1 to 4)	109, 62, 41, 24
Busiest hour	30 commits
Pull requests open at once	5.8 on average
Pull request open to merge	Median 20 min, 75th percentile 198 min, 90th percentile 525 min
Tier 1 on a warm machine	About 55 to 64 seconds
Tier 1 and Tier 2 on the runner	257 seconds and 8.4 minutes
Lander lock held	About 1 minute (6% busy at 3.8 an hour, about 50% at the peak)
Changes touching no code or spec	35 of 58
Changes over 150 lines	15 of 58

Table 5. What lung measures. The wait is for the lander’s turn, not for the checks.

ELEMENT	STATUS	WHAT LUNG HAS, OR WHAT IT LACKS AND WHY
Which design (A or B)	Keep A	The serial lander under a file lock plus a merge pinned to the head commit is a hand-made queue. Lock time times rate is 6%, so design B buys nothing yet. Write down the trigger: more than 60 a day sustained and more than 50% lock use
Certificate	Missing	No certificate and no Certificate-Base line (0 of 60). Task-ID is on 57 of 60, Agent-Run on 56, Feature-Flag on 54, and conventional titles on 60. No hook checks any of them
Overlap check	Missing	The lander always rebases and runs the full Tier 1. It would gain most on docs, scripts and spec-only changes (60% of changes). Edits to the agent code touch most modules (96 of 112), so the overlap there is seldom empty
Reservations	Missing	Do commuting first. 111 of 237 concurrent pairs share a file; removing five shared edit points leaves 36 (65% fewer). File-level locks are enough; function-level is not worth it
Edits that commute	Missing	The test list in tier1.sh, the check list in Check.lean, the spec umbrella file and two bookkeeping files are edited by nearly every change. Generate each from a sorted listing, with a freshness check and a failing self-test
Single writer	Partial	Has one lock, a scratch worktree, rebase, ratchet, Tier 1, a lease push, retry up to 3 times. Lacks the certificate check, the shortcut, freeze, trailer, size and time checks, and failures returned as data. A lander agent runs it, hence the 198-minute tail
Freeze on red	Missing	Only an opt-in halt that nobody sets. An open pull request adds a check, but its exemption is a title anyone can write, and it would freeze everything while main is red for a false reason
Revert watchdog	Live, misfiring	Merged on Nov 4. It reverted a good commit because Tier 2 failed on a shallow checkout, could not open its pull request, and left 12 stray branches and 18 open issues
Switches and replace in steps	Have, different	Lung uses reachability: a spec module lands unwired and a later change wires it, checked mechanically. Better than a runtime switch for a verified core. Missing: a check that a change adding behaviour says so
Commit lines	Have, no check	Titles follow the rule 60 of 60; nothing enforces it. About 25 lines of checking would do
Time and size limits	Missing	Only words in AGENTS.md. The 150-line rule is broken 26% of the time, so make it a check with a waiver line, tightened over time. The time limit belongs to the task manager
Repeatable results	Have (shell), missing (Lean)	Shell lints exist with self-tests. Missing: unordered directory reads and hash-map loops in Lean, and a rerun-and-compare for flaky tests
Lander as spec	Missing	No landing module in the spec. Wanted: states, steps and theorems (lands only when base is head or the overlap check passed; frozen admits only revert or fix; one change per step), then a small Lean program replaces the shell policy

Table 6. Lung against the design: have and missing. Source: the delta map for lung.

SLICE	WHAT	NEEDS
S0	Fix the false reds in Tier 2 (full checkout, watchdog skips infrastructure failures). About 10 lines	Nothing; gate for S1 and trust in main
S1	Land the open freeze check and fix its exemption. About 20 lines	S0
S2	Make edits commute: four independent changes (test list, check list, spec umbrella, bookkeeping files)	Nothing
S3	Certificate: write and verify. About 90 lines	S0 for trust in main
S4	Lander shortcut for empty change, then the overlap selection. About 40 then 110 lines	S3
S5	Land-time checks: commit lines, size, real Agent-Run. About 70 lines	Nothing
S6	Repeatability: Lean directory and hash-map lint, rerun-and-compare. About 60 lines	Nothing
S7	Lander as Lean spec, then the decision program replaces the shell policy	S1 and S4
S8	Lander as a service that picks up green pull requests, removing the long tail	S1 and S4

Table 7. The ordered slices. Source: the delta map for lung.

Questions for the owner

1. Stay on design A? **Recommended: yes**, and record the trigger for B.
2. Should Tier 2 repeat Tier 1? **No.** It costs 4.3 of its 8.4 minutes; run the build and ratchets only.
3. Sign certificates? **Not yet.** Everything runs as one user, so a signature adds nothing; bind to the tree and revisit with a second host.
4. How is a fix to a red main let through? **Only if the change touches nothing outside tests and CI files**; a title is forgeable.
5. Does a Tier 2 failure that also fails on the parent get reverted? **No.** Open an issue; no revert and no freeze beyond one run.
6. May the spec umbrella file be generated without the owner? **Yes**, if a check proves it is exactly the sorted module list.
7. Is the 150-line limit hard? **No:** tighten gradually, with a waiver line.

9 Appendix A: the AGENTS.md text

Place this at the root of every repository. Agents read it when they start. It is the short form; the design note's longer form adds the same rules with a few more details.

```
# Autonomous Trunk-Based Development Protocol

## 1. Workspace & Branch Lifecycle
- Always start from a fresh synchronization of trunk: `git fetch origin && git rebase
origin/main`.
- Operate strictly within an isolated ephemeral Git worktree per task (`git worktree add
../wt-<task-id> -b agent/<task-id> origin/main`).
- Scope limit: Diffs must not exceed 150 lines of production code. If a task requires
larger changes, halt execution and output a decomposition plan using Branch by Abstraction.
- Hard TTL: Any worktree older than 45 minutes without a passing build must be aborted and
decomposed.

## 2. Defensive Coding Mandates (Zero Broken Trunk)
- Every commit must compile cleanly with zero warnings and pass all Tier 1 pre-flight
checks.
- Incomplete features, new endpoints, or behavioral changes MUST be gated behind a
programmatic feature flag defaulting to `false`.
- Breaking changes to shared interfaces or schemas are forbidden. Use the Expand-and-
Contract pattern:
  1. Add new interface/field alongside legacy code.
  2. Migrate callers incrementally across micro-commits.
  3. Remove legacy code only when zero references remain.
- Enforce strict determinism: Sort all listings, map keys, and iteration paths.

## 3. Pre-Landing Verification & Base Certificate
Before submitting code to the Delta Lander:
1. Rebase onto latest trunk: `git fetch origin main && git rebase origin/main`.
2. Execute fast static analysis, type checking, and affected unit tests (Tier 1 <60s).
3. Attach the Base Certificate trailer to your commit message:

feat(auth): implement token refresh retry mechanism

Task-ID: TASK-1024 Certificate-Base: 4f8b9a2c... Certificate-Status: PASS

## 4. Failure & Rollback Protocol
- Never force-push directly to `main`.
- If a commit breaks post-merge verification, the system will automatically revert it. Do
not attempt hotfixes directly on trunk; pull the revert, reproduce the failure in a new
ephemeral worktree, and re-submit.
```

10 Appendix B: the lander sketch, with review notes

A GitHub Action for the lander: it reads the certificate from the commit, compares main with the claimed base, lands at once when main has not moved, and otherwise runs the checks that touch the overlap. It is a sketch to learn from, not a drop-in. The review notes follow it.

```yaml
name: Agentic Delta Lander

on:
  pull_request:
    types: [opened, synchronize, labeled]

jobs:
  delta-evaluator:
    runs-on: ubuntu-latest
    permissions:
      contents: write
      pull-requests: write

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4
        with:
          fetch-depth: 0

      - name: Verify Main Branch Health (Freeze-on-Red Gate)
        run: |
          MAIN_STATUS=$(gh run list --branch main --workflow "Continuous Integration" --
limit 1 --json conclusion -q '..conclusion')
          if [ "$MAIN_STATUS" = "failure" ]; then
            echo "::error::Main branch is currently RED. Merges frozen until baseline is
restored."
            exit 1
          fi
        env:
          GH_TOKEN: ${{ secrets.GITHUB_TOKEN }}

      - name: Extract Base Certificate
        id: cert
        run: |
          PR_HEAD_SHA="${{ github.event.pull_request.head.sha }}"
          BASE_SHA=$(git log -1 --format="%(trailers:key=Certificate-Base,valueonly=true)"
$PR_HEAD_SHA | tr -d ' ')
          STATUS=$(git log -1 --format="%(trailers:key=Certificate-Status,valueonly=true)"
$PR_HEAD_SHA | tr -d ' ')

          if [ -z "$BASE_SHA" ] || [ "$STATUS" != "PASS" ]; then
            echo "::error::Missing or invalid Base Certificate trailer (Certificate-Base /
Certificate-Status)."
            exit 1
          fi

          echo "base_sha=$BASE_SHA" >> $GITHUB_OUTPUT
          echo "head_sha=$PR_HEAD_SHA" >> $GITHUB_OUTPUT

      - name: Compute Delta (main vs. Base SHA)
        id: delta
        run: |
          BASE_SHA="${{ steps.cert.outputs.base_sha }}"
          CURRENT_MAIN_SHA=$(git rev-parse origin/main)
```

```
          if [ "$BASE_SHA" = "$CURRENT_MAIN_SHA" ]; then
            echo "delta_type=empty" >> $GITHUB_OUTPUT
            echo "Empty delta detected (main has not moved past base SHA $BASE_SHA)."
          else
            echo "delta_type=non_empty" >> $GITHUB_OUTPUT
            echo "Non-empty delta detected between $BASE_SHA and $CURRENT_MAIN_SHA."
          fi

      - name: Execute Fast-Forward Merge (Empty Delta)
        if: steps.delta.outputs.delta_type == 'empty'
        run: |
          echo "Landing submission via immediate fast-forward merge..."
          git checkout main
          git merge --ff-only ${{ steps.cert.outputs.head_sha }}
          git push origin main

      - name: Evaluate Interaction Suite (Non-Empty Delta)
        if: steps.delta.outputs.delta_type == 'non_empty'
        run: |
          BASE_SHA="${{ steps.cert.outputs.base_sha }}"
          HEAD_SHA="${{ steps.cert.outputs.head_sha }}"

          echo "Finding overlapping changed files between PR and delta on main..."
          # List files changed in PR
          PR_FILES=$(git diff --name-only $BASE_SHA $HEAD_SHA)
          # List files changed on main since base
          MAIN_FILES=$(git diff --name-only $BASE_SHA origin/main)

          OVERLAP=$(comm -12 <(echo "$PR_FILES" | sort) <(echo "$MAIN_FILES" | sort))

          if [ -n "$OVERLAP" ]; then
            echo "::warning::Direct file overlap detected in delta:"
            echo "$OVERLAP"
          fi

          echo "Running targeted interaction test suite and determinism lints..."
          # Execute fast Tier 1 static checks and tests touching affected files
          npm run lint:determinism || cargo test --partition affected

          echo "Speculatively merging and landing onto main..."
          git checkout main
          git merge --no-ff ${{ steps.cert.outputs.head_sha }} -m "merge: agent submission
[delta verified]"
          git push origin main
```

#	NOTE	WHAT TO CHANGE
1	One writer at a time	Two runs can land at once and race on main. Put the job in a single concurrency group (for example `group: lander`) with `cancel-in-progress: false`, so runs queue and none is dropped
2	Pushes by the built-in token start no workflows	A push made with `GITHUB_TOKEN` does not trigger other workflows, so Tier 2 would never run on a landed change. The exemption is `workflow_dispatch`: have the lander start Tier 2 explicitly with `gh workflow run`
3	The freeze check has a bug	The query `'..conclusion'` is not valid for the first run's result. It should read the first item (`.[0].conclusion`). As written the gate can fail open, so a red main would not freeze
4	The interaction step is a stub	`npm run lint:determinism || cargo test ...` runs one or the other, and a failed lint is hidden when the second succeeds. Run both, require both, and select tests from the overlap
5	No merge commits	The non-empty road uses `git merge --no-ff`, which makes a merge commit and breaks the design's fast-forward-only rule. Rebase on the worker side, re-certify, and land with `--ff-only`
6	The certificate asserts itself	The worker wrote the trailer, so nothing proves the tests ran. It is an integrity check, not a proof. Bind it to the diff hash and tree, and sign it once a second user or host exists
7	Name the head commit	Use the exact head commit of the pull request throughout: check out that commit and merge it by hash, and stop if the head moved since the run began, so a late push cannot slip in

Table 8. Review notes on the lander sketch.